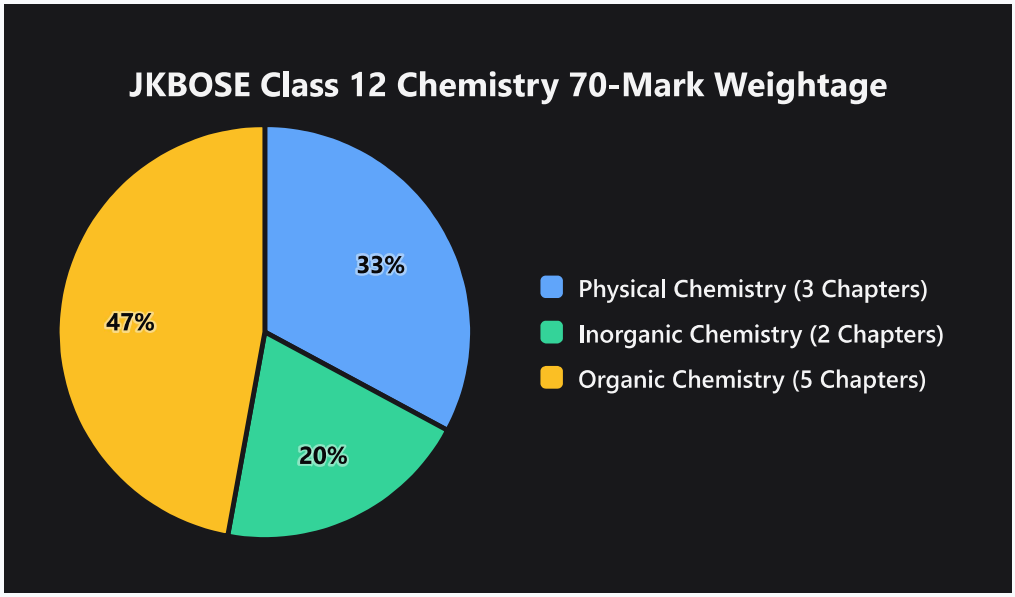
JKBOSE Class 12 Chemistry: Ultimate Master Blueprint (70/70 Hard-Paper Shield)

> **Target**: 70/70 in Theory (Total 100 with 30 Practicals)
> **Board**: Jammu & Kashmir Board of School Education (JKBOSE)
> **Syllabus**: NCERT Rationalized 10-Chapter Curriculum
> **Design Philosophy**: Built with "Hard Paper Shock-Absorbers" — guarantees 95%+ even if the board sets tricky conceptual MCQs, multi-step numericals, or unconventional organic conversions.

0. The Complete 70-Mark Architecture & Strategic Allocation



Section / Branch	Chapter Name	Weightage	Question Types Dominating the Paper
Physical (23 Marks)	1. Solutions	**7 Marks**	3M Numerical (Colligative/Van't Hoff) + 2M Raoult/Henry Law + 2M Ideal/Non-ideal
	2. Electrochemistry	**9 Marks**	5M Master Derivation / Nernst Equation + 3M Kohlrausch/Conductivity + 1M MCQ
	3. Chemical Kinetics	**7 Marks**	5M Integrated Rate / Arrhenius Derivation + 2M Order/Molecularity / Rate Law
Inorganic (14 Marks)	4. d- & f-Block Elements	**7 Marks**	3M Scientific Reasoning + 2M Lanthanoid Contraction + 2M $KMnO_4/K_2Cr_2O_7$ KMnO4/K2Cr2O7
	5. Coordination Compounds	**7 Marks**	3M VBT / CFT Splitting + 2M IUPAC Nomenclature + 2M Isomerism
Organic (33 Marks)	6. Haloalkanes & Haloarenes	**6 Marks**	3M S_N1 SN1 vs S_N2 SN2 Mechanism + 2M Named Reaction / Elimination + 1M MCQ
	7. Alcohols, Phenols & Ethers	**6 Marks**	3M Mechanisms (Hydration/Dehydration) + 2M Reimer-Tiemann/Kolbe + 1M Test
	8. Aldehydes, Ketones, Acids	**8 Marks**	5M Long Question (Aldol/Cannizzaro/Mechanism) + 3M Conversions/Chemical Tests
	9. Amines	**6 Marks**	3M Basicity Order Reasoning + 2M Hoffmann/Carbylamine + 1M Test
	10. Biomolecules	**7 Marks**	3M Protein Denaturation/Structure + 2M DNA vs RNA + 2M Glucose Reactions/Vitamins
Grand Total	**10 Chapters**	**70 Marks**	**Time: 3 Hours (Theory)**

PART 1: PHYSICAL CHEMISTRY (23 / 23 MARKS)

Physical Chemistry is purely mathematical and law-based. Every single mark can be secured by mastering 4 core derivations, 4 laws, and 5 numerical templates.

1.1 The Guaranteed Physical Derivations (5 & 3 Markers)

Derivation 1: Integrated Rate Law for First-Order Reaction & Half-Life

- **Reaction**: $R \rightarrow P$R → P

- **Differential Rate Expression**:

$$-\frac{d[R]}{dt} = k[R]^1 \implies \frac{d[R]}{[R]} = -kdt$$

−dtd[R] = k[R]1 ⟹ [R]d[R] = −k dt

- **Integration**:

$$\int \frac{d[R]}{[R]} = -k\int dt \implies \ln[R] = -kt + I \quad \text{— (Eq 1)}$$

∫ [R]d[R] = −k ∫ dt ⟹ ln[R] = −kt + I — (Eq 1)

- **Boundary Condition ($t = 0$t = 0)**: At $t = 0$t = 0, $[R] = [R]_0 \implies \ln[R]_0 = 0 + I \implies I = \ln[R]_0$ [R] = [R]0 ⟹ ln[R]0 = 0 + I ⟹ I = ln[R]0.

- **Rate Constant Equation**:

$$\ln[R] = -kt + \ln[R]_0 \implies kt = \ln\frac{[R]_0}{[R]}$$

ln[R] = −kt + ln[R]0 ⟹ kt = ln [R][R]0

$$k = \frac{2.303}{t}\log_{10}\frac{[R]_0}{[R]} \quad \text{or} \quad t = \frac{2.303}{k}\log_{10}\frac{[R]_0}{[R]}$$

k = t2.303 log10 [R][R]0 or t = k2.303 log10 [R][R]0

- **Half-Life Period ($t_{1/2}$ t1/2)**: At $t = t_{1/2}$ t = t1/2, $[R] = \frac{[R]_0}{2}$[R] = 2[R]0:

$$t_{1/2} = \frac{2.303}{k}\log_{10}(\frac{[R]_0}{[R]_0/2}) = \frac{2.303 \times \log_{10}(2)}{k} = \frac{2.303 \times 0.3010}{k} = \frac{0.693}{k}$$

t1/2 = k2.303 log10 ([R]0/2[R]0) = k2.303 × log10(2) = k2.303 × 0.3010 = k0.693

- *Hard Paper Shock-Absorber*: If asked for **Zero-Order Reaction**:

$$-\frac{d[R]}{dt} = k[R]^0 = k \implies [R] = -kt + [R]_0 \implies k = \frac{[R]_0 - [R]}{t}, \quad t_{1/2} = \frac{[R]_0}{2k}$$

−dtd[R] = k[R]0 = k ⟹ [R] = −kt + [R]0 ⟹ k = t[R]0 − [R], t1/2 = 2k[R]0

Derivation 2: Arrhenius Equation (Temperature Dependence of Rate Constant)

- **Arrhenius Equation**: $k = Ae^{-E_a/RT}$ k = Ae−Ea/RT
- Taking natural logarithm on both sides:

$$\ln k = \ln A - \frac{E_a}{RT} \implies \log_{10} k = \log_{10} A - \frac{E_a}{2.303RT}$$

ln k = ln A − RT Ea ⟹ log10 k = log10 A − 2.303RT Ea*(A plot of* $\log k$*log k vs* $\frac{1}{T}$*T 1 gives a straight line with slope* $= -\frac{E_a}{2.303R}$*= −2.303REa and intercept* $= \log A$*= log A).*

- At temperature T_1 T1, rate constant is k_1 k1: $\ln k_1 = \ln A - \frac{E_a}{RT_1}$ln k1 = ln A − RT1 Ea.
- At temperature T_2 T2, rate constant is k_2 k2: $\ln k_2 = \ln A - \frac{E_a}{RT_2}$ln k2 = ln A − RT2 Ea.
- Subtracting equation 1 from equation 2:

$$\ln\frac{k_2}{k_1} = \frac{E_a}{R}(\frac{1}{T_1} - \frac{1}{T_2}) \implies \log_{10}\frac{k_2}{k_1} = \frac{E_a}{2.303R}[\frac{T_2 - T_1}{T_1 T_2}]$$

ln k1k2 = REa (T11 − T21) ⟹ log10 k1k2 = 2.303REa [T1T2T2 − T1]

Derivation 3: Colligative Properties — Depression in Freezing Point (ΔT_f ΔTf) & Elevation in Boiling Point (ΔT_b ΔTb)

- **Depression in Freezing Point**:
  $$\Delta T_f = T_f^\circ - T_f \propto m \Longrightarrow \Delta T_f = K_f \cdot m$$
  ΔTf = Tf∘ − Tf ∝ m ⟹ ΔTf = Kf · m *(where K_f Kf is molal depression constant / cryoscopic constant, defined as the depression in freezing point for a 1 molal solution)*. Molality $m = \frac{\text{moles of solute}}{\text{mass of solvent in kg}} = \frac{w_2/M_2}{w_1/1000} = \frac{w_2 \times 1000}{M_2 \times w_1}$m = mass of solvent in kgmoles of solute = w1/1000w2/M2 = M2 ×w1w2×1000.
  $$\Delta T_f = K_f \times \frac{w_2 \times 1000}{M_2 \times w_1} \Longrightarrow M_2 = \frac{K_f \times w_2 \times 1000}{\Delta T_f \times w_1}$$
  ΔTf = Kf × M2 × w1w2 × 1000 ⟹ M2 = ΔTf × w1Kf × w2 × 1000
- **Elevation in Boiling Point**:
  $$\Delta T_b = T_b - T_b^\circ = K_b \cdot m \Longrightarrow M_2 = \frac{K_b \times w_2 \times 1000}{\Delta T_b \times w_1}$$
  ΔTb = Tb − Tb∘ = Kb · m ⟹ M2 = ΔTb × w1Kb × w2 × 1000
- **Osmotic Pressure (ΠΠ)**:
  $$\Pi = CRT = \frac{n_2}{V}RT = \frac{w_2\,RT}{M_2\,V} \Longrightarrow M_2 = \frac{w_2\,RT}{\Pi V}$$
  Π = CRT = V n2RT = M2V w2RT ⟹ M2 = ΠVw2RT*(Why Osmotic Pressure is best for determining molar masses of polymers and biomolecules: Measured at room temperature, uses molarity instead of molality, and magnitudes are large enough to be read accurately)*.

Derivation 4: Van 't Hoff Factor (ii) and Degree of Dissociation / Association

$$i = \frac{\text{Normal Molar Mass}}{\text{Abnormal (Observed) Molar Mass}} = \frac{\text{Observed Colligative Property}}{\text{Calculated Colligative Property}} = \frac{\text{Total particles after association/dissociation}}{\text{Total particles before}}$$

i = Abnormal (Observed) Molar MassNormal Molar Mass

= Calculated Colligative PropertyObserved Colligative Property

= Total particles beforeTotal particles after association/dissociation

- **Degree of Dissociation (αα)**: For an electrolyte $A_n \rightarrow nA$An → nA: initial moles $= 1$= 1, at equilibrium moles $= 1 - \alpha + n\alpha$= 1 − α + nα.
  $$i = \frac{1 + (n-1)\alpha}{1} \Longrightarrow \alpha = \frac{i-1}{n-1}$$
  i = 11 + (n − 1)α ⟹ α = n − 1i − 1
- **Degree of Association (αα)**: For dimerization/association $nA \rightarrow A_n$ nA → An: initial moles $= 1$= 1, at equilibrium moles $= 1 - \alpha + \frac{\alpha}{n}$= 1 − α + nα.
  $$i = 1 - \alpha(1 - \frac{1}{n}) \Longrightarrow \alpha = \frac{1-i}{1-1/n}$$
  i = 1 − α(1 − n1) ⟹ α = 1 − 1/n1 − i

1.2 Fundamental Laws & Conceptual Rules (Physical)

1. **Kohlrausch's Law of Independent Migration of Ions**:

> *"At infinite dilution, when dissociation of electrolyte is complete, each ion makes a definite contribution towards molar conductivity of electrolyte irrespective of nature of other ion with which it is associated."*
>
> $$\Lambda_m^\circ = \nu_+ \lambda_+^\circ + \nu_- \lambda_-^\circ$$
>
> Λm∘ = ν+λ+∘ + ν−λ−∘*Applications*:

- Calculate Λ_m° Λm∘ for weak electrolytes:
  $\Lambda_m^\circ (CH_3COOH) = \Lambda_m^\circ (CH_3COONa) + \Lambda_m^\circ (HCl) - \Lambda_m^\circ (NaCl)$Λm∘(CH3COOH) = Λm∘(CH3COONa) + Λm∘(HCl) − Λm∘(NaCl).
- Determine degree of dissociation ($\alpha = \frac{\Lambda_m}{\Lambda_m^\circ}$α = Λm∘Λm) and dissociation constant ($K_a = \frac{c\alpha^2}{1-\alpha}$Ka = 1−αcα2).

2. **Raoult's Law**:
   - For volatile liquids: $p_A = p_A^\circ x_A$ pA = pA∘xA and $p_B = p_B^\circ x_B$ pB = pB∘xB; $p_{total} = p_A + p_B$ ptotal = pA + pB.
   - For non-volatile solute: Relative lowering of vapour pressure $= \frac{p_A^\circ - p_A}{p_A^\circ} = x_B = \frac{n_B}{n_A + n_B}$= pA∘pA∘−pA = xB = nA+nBnB.
3. **Ideal vs Non-Ideal Solutions**:

Property	Ideal Solutions	Positive Deviation (Non-Ideal)	Negative Deviation (Non-Ideal)
Intermolecular Forces	$A-B \approx A-A \approx B-B$ A − B ≈ A − A ≈ B − B	$A-B < A-A$ or $B-B$ A − B < A − A or B − B (weaker)	$A-B > A-A$ or $B-B$ A − B > A − A or B − B (stronger)
ΔH_{mixing} **ΔHmixing**	$= 0$= 0	> 0> 0 (Endothermic)	< 0< 0 (Exothermic)
ΔV_{mixing} **ΔVmixing**	$= 0$= 0	> 0> 0 (Expansion)	< 0< 0 (Contraction)
Vapour Pressure	Obeys Raoult's law	Higher than expected	Lower than expected
Azeotrope Type	No azeotrope	**Minimum boiling azeotrope**	**Maximum boiling azeotrope**
Examples	nn-Hexane + nn-Heptane, Benzene + Toluene	Ethanol + Acetone, CS_2 CS2 + Acetone	Acetone + Chloroform, $HNO_3 + H_2O$HNO3 + H2O

4. **Henry's Law**: $p = K_H \cdot x$p = KH · x (Partial pressure of gas in vapour phase is proportional to mole fraction of gas in solution).
   - Higher the value of K_H KH at a given pressure, the **lower the solubility** of the gas in liquid.

- K_H KH increases with temperature, hence gases are **less soluble at higher temperatures** (why aquatic species are more comfortable in cold water).

1.3 Master Physical Chemistry Numerical Templates

Template 1: Nernst Equation & Gibbs Free Energy (Electrochemistry)

- **General Cell Reaction**: $aA + bB \rightarrow cC + dD$aA + bB → cC + dD

$$E_{cell} = E^\circ_{cell} - \frac{2.303RT}{nF}\log_{10} Q = E^\circ_{cell} - \frac{0.0591}{n}\log_{10}\frac{[C]^c\,[D]^d}{[A]^a\,[B]^b} \quad \text{(at 298 K)}$$

Ecell = Ecell∘ − nF2.303RT log10 Q = Ecell∘ − n0.0591 log10 [A]a[B]b[C]c[D]d (at 298 K) *(where nn is number of electrons exchanged, $E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$ Ecell∘ = Ecathode∘ − Eanode∘).*

- **Standard Free Energy & Equilibrium Constant**:

$$\Delta G^\circ = -nFE^\circ_{cell}, \qquad \log_{10} K_c = \frac{nE^\circ_{cell}}{0.0591}$$

ΔG∘ = −nFEcell∘, log10 Kc = 0.0591nEcell∘

- *Standard Example*: Calculate EMF of cell $\mathrm{Mg(s)|Mg^{2+}\,(0.001\ M)||Cu^{2+}\,(0.0001\ M)|Cu(s)}$Mg(s)|Mg2+ (0.001 M)||Cu2+(0.0001 M)|Cu(s).
$n = 2$n = 2, $E^\circ_{cell} = 0.34 - (-2.37) = +2.71$ VEcell∘ = 0.34 − (−2.37) = +2.71 V.
$E_{cell} = 2.71 - \frac{0.0591}{2}\log\frac{10^{-3}}{10^{-4}} = 2.71 - 0.02955(1) = 2.68$ VEcell = 2.71 − 20.0591 log 10−410−3 = 2.71 − 0.02955(1) = 2.68 V.

Template 2: Molar Conductivity & Cell Constant

- Cell constant $G^* = \frac{l}{A} = R \times \kappa$G∗ = Al = R × κ.
- Conductivity $\kappa = \frac{G^*}{R}$κ = RG∗.
- Molar Conductivity $\Lambda_m = \frac{\kappa \times 1000}{M}$Λm = Mκ×1000 (where κκ is in $\mathrm{S \cdot cm^{-1}}$ S · cm−1 and MM is molarity).

Template 3: First-Order Reaction Timing Ratios

- **Show that for a first-order reaction, time required for 99%99% completion is double the time required for 90%90% completion**:
  1. $t_{99\%} = \frac{2.303}{k}\log\frac{100}{100-99} = \frac{2.303}{k}\log(100) = \frac{2.303}{k} \times 2 = \frac{4.606}{k}$t99% = k2.303 log 100−99100 = k2.303 log(100) = k2.303 × 2 = k4.606.
  2. $t_{90\%} = \frac{2.303}{k}\log\frac{100}{100-90} = \frac{2.303}{k}\log(10) = \frac{2.303}{k} \times 1$t90% = k2.303 log 100−90100 = k2.303 log(10) = k2.303 × 1.
  3. $\frac{t_{99\%}}{t_{90\%}} = \frac{4.606/k}{2.303/k} = 2$t90%t99% = 2.303/k4.606/k = 2. Hence, $t_{99\%} = 2 \times t_{90\%}$ t99% = 2 × t90%. (Q.E.D.)

PART 2: INORGANIC CHEMISTRY (14 / 14 MARKS)

Inorganic Chemistry is heavily based on reasoning patterns, electronic configurations, and coordination principles.

2.1 Unit 4: d- and f-Block Elements (7 Marks)

The 10 Irrefutable Scientific Reasoning Answers

1. **Why do transition elements exhibit variable oxidation states?**
   Answer: Because the energy levels of $(n-1)d$(n − 1)d and nsns orbitals are very close to each other. Both nsns and unpaired $(n-1)d$(n − 1)d electrons can participate in bond formation.
2. **Why are transition metal ions colored in aqueous solutions?**
   Answer: Due to $d-d$**d − d electron transitions**. In the presence of water/ligands, the degenerate dd-orbitals split into two sets (t_{2g} t2g and e_g eg). Electrons absorb light from the visible region and get promoted to higher dd-orbitals, transmitting complementary colors.
   (Note: Sc^{3+} (d^0)*Sc3+(d0),* Ti^{4+} (d^0)*Ti4+(d0), and* Zn^{2+} (d^{10})*Zn2+(d10) are* ***colorless*** *because they have no unpaired* d*d-electrons for* $d-d$*d − d transition).*
3. **Why do transition metals act as excellent catalysts?**
   Answer: (i) Ability to adopt **variable oxidation states** and form unstable intermediate complexes, and (ii) Large surface area providing active sites to adsorb reactant molecules, lowering activation energy.
4. **Why do transition elements have high melting points and high enthalpies of atomization?**
   Answer: Due to strong interatomic metallic bonding involving both nsns and unpaired $(n-1)d$(n − 1)d electrons. Greater the number of unpaired dd-electrons, stronger is the metallic bonding.
5. **What are Interstitial Compounds, and what are their properties?**
   Answer: Compounds formed when small non-metal atoms (like HH, CC, NN, BB) get trapped inside the interstitial spaces of the transition metal crystal lattice (e.g., TiCTiC, $\mathrm{Fe_3H}$Fe3H).
   Properties: (i) Very high melting points (higher than pure metals), (ii) Extremely hard, (iii) Retain metallic conductivity, (iv) Chemically inert.
6. **What is Lanthanoid Contraction? What are its causes and consequences?**
   Answer:
   Definition: The steady and gradual decrease in the atomic and ionic radii of lanthanoid elements with increasing atomic number from La^{3+} La3+ to Lu^{3+} Lu3+.
   Cause: The addition of electrons to the inner $4f$4f subshell. The $4f$4f electrons have poor, diffuse shielding shapes and fail to shield the increasing nuclear charge effectively, causing the outer electrons to experience an increased effective nuclear charge (Z_{eff} Zeff).
   Consequences:
   (a) **Identical radii of** $4d$**4d and** $5d$**5d elements**: Elements of $4d$4d and $5d$5d series belonging to the same group have almost identical atomic sizes (e.g., $\mathrm{Zr}\approx\mathrm{Hf}$ Zr ≈ Hf, $\mathrm{Nb}\approx\mathrm{Ta}$ Nb ≈ Ta).

(b) **Difficulty in separation**: Due to identical chemical properties and sizes, separation of lanthanoids is extremely difficult.
(c) **Decrease in basic strength**: Basic strength of hydroxides decreases from $\mathrm{La(OH)_3}$ La(OH)3 (most basic) to $\mathrm{Lu(OH)_3}$ Lu(OH)3 (least basic) because covalent character increases with decreasing ionic size.

7. **Why is $\mathrm{Cr^{2+}}$ Cr2+ reducing while $\mathrm{Mn^{3+}}$ Mn3+ is oxidizing, though both have d^4 d4 configuration?**
Answer: $\mathrm{Cr^{2+}}$ Cr2+ loses an electron to form $\mathrm{Cr^{3+}}$ Cr3+, which has a stable, half-filled t_{2g}^3 t2g3 configuration in aqueous medium. Conversely, $\mathrm{Mn^{3+}}$ Mn3+ gains an electron to form $\mathrm{Mn^{2+}}$ Mn2+, achieving a stable half-filled d^5 d5 configuration.
8. **Why is $E^\circ_{\mathrm{Cu^{2+}/Cu}}$ ECu2+/Cu∘ positive ($+0.34\ \mathrm{V}$ +0.34 V)?**
Answer: The high enthalpy of atomization and high ionization energies ($\mathrm{IE_1 + IE_2}$ IE1 + IE2) of Copper are not balanced by its enthalpy of hydration.
9. **Why are ZnZn, CdCd, and HgHg not considered typical transition elements?**
Answer: Because they have fully filled dd-subshells ($(n-1)d^{10}$ (n − 1)d10) in their ground atomic state as well as in their common oxidation states ($\mathrm{Zn^{2+}}$ Zn2+, $\mathrm{Cd^{2+}}$ Cd2+, $\mathrm{Hg^{2+}}$ Hg2+).
10. **Oxidizing properties of $\mathrm{KMnO_4}$ KMnO4 and $\mathrm{K_2Cr_2O_7}$ K2Cr2O7 in Acidic Medium**:
    - $\mathrm{K_2Cr_2O_7}$ K2Cr2O7: $\mathrm{Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O}$ Cr2O72− + 14H+ + 6e− → 2Cr3+ + 7H2O
        - Oxidizes $\mathrm{Fe^{2+} \rightarrow Fe^{3+}}$ Fe2+ → Fe3+: $\mathrm{Cr_2O_7^{2-} + 14H^+ + 6Fe^{2+} \rightarrow 2Cr^{3+} + 6Fe^{3+} + 7H_2O}$ Cr2O72− + 14H+ + 6Fe2+ → 2Cr3+ + 6Fe3+ + 7H2O
        - Oxidizes $\mathrm{I^- \rightarrow I_2}$ I− → I2: $\mathrm{Cr_2O_7^{2-} + 14H^+ + 6I^- \rightarrow 2Cr^{3+} + 3I_2 + 7H_2O}$ Cr2O72− + 14H+ + 6I− → 2Cr3+ + 3I2 + 7H2O
    - $\mathrm{KMnO_4}$ KMnO4: $\mathrm{MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O}$ MnO4− + 8H+ + 5e− → Mn2+ + 4H2O
        - Oxidizes Oxalate ($\mathrm{C_2O_4^{2-} \rightarrow CO_2}$ C2O42− → CO2): $\mathrm{2MnO_4^- + 16H^+ + 5C_2O_4^{2-} \rightarrow 2Mn^{2+} + 10CO_2 + 8H_2O}$ 2MnO4− + 16H+ + 5C2O42− → 2Mn2+ + 10CO2 + 8H2O

2.2 Unit 5: Coordination Compounds (7 Marks)

A. IUPAC Nomenclature Mastery

- **Rules**: Cation named first; ligands in alphabetical order with prefixes di-, tri-; oxidation state in Roman numerals in parentheses; if complex ion is an anion, metal ends in **-ate** (e.g., Ferrate, Cobaltate, Platinate).

1. $\mathrm{[Co(NH_3)_5Cl]Cl_2}$ [Co(NH3)5Cl]Cl2: **Pentaamminechloridocobalt(III) chloride**
2. $\mathrm{K_3[Fe(CN)_6]}$ K3[Fe(CN)6]: **Potassium hexacyanidoferrate(III)**
3. $\mathrm{[Cr(en)_3]Cl_3}$ [Cr(en)3]Cl3: **Tris(ethane-1,2-diamine)chromium(III) chloride**
4. $\mathrm{[Pt(NH_3)_2Cl(NO_2)]}$ [Pt(NH3)2Cl(NO2)]: **Diamminechloridonitrito-N-platinum(II)**

5. $K_2[Ni(CN)_4]$K2[Ni(CN)4]: **Potassium tetracyanidonickelate(II)**

B. Valence Bond Theory (VBT) Complex Predictions

Complex	Central Ion	Configuration	Ligand Type	Hybridization	Geometry	Magnetic Property	Spin Type
$[Fe(CN)_6]^{3-}$ [Fe(CN)6]3−	Fe^{3+} Fe3+ (d^5 d5)	t_{2g}^5 t2g5 (forced pairing)	Strong (CN^- CN−)	d^2sp^3 d2sp3	Octahedral	Paramagnetic ($n = 1$n = 1)	Low Spin (Inner)
$[FeF_6]^{3-}$ [FeF6]3−	Fe^{3+} Fe3+ (d^5 d5)	$e_g^2 t_{2g}^3$ eg2t2g3 (no pairing)	Weak (F^- F−)	sp^3d^2 sp3d2	Octahedral	Paramagnetic ($n = 5$n = 5)	High Spin (Outer)
$[Ni(CN)_4]^{2-}$ [Ni(CN)4]2−	Ni^{2+} Ni2+ (d^8 d8)	d^8 d8 paired	Strong (CN^- CN−)	dsp^2 dsp2	Square Planar	Diamagnetic ($n = 0$n = 0)	Low Spin
$[NiCl_4]^{2-}$ [NiCl4]2−	Ni^{2+} Ni2+ (d^8 d8)	d^8 d8 unpaired (22)	Weak (Cl^- Cl−)	sp^3 sp3	Tetrahedral	Paramagnetic ($n = 2$n = 2)	High Spin
$[Co(NH_3)_6]^{3+}$ [Co(NH3)6]3+	Co^{3+} Co3+ (d^6 d6)	t_{2g}^6 t2g6 (all paired)	Strong (NH_3 NH3)	d^2sp^3 d2sp3	Octahedral	Diamagnetic ($n = 0$n = 0)	Low Spin (Inner)

Magnetic moment formula: $\mu = \sqrt{n(n+2)}$ BMμ = n(n + 2)

BM (where nn is number of unpaired electrons).

C. Crystal Field Theory (CFT)

- In an octahedral field, spherical dd-orbitals split into:
  - Higher energy e_g **eg** set ($d_{x^2-y^2}$, d_{z^2} dx2−y2, dz2) raised by $+0.6\Delta_o$ +0.6Δo.
  - Lower energy t_{2g} **t2g** set (d_{xy}, d_{yz}, d_{xz} dxy, dyz, dxz) lowered by $-0.4\Delta_o$ −0.4Δo.
- **Condition for High-Spin vs Low-Spin**:
  - If $\Delta_o < P$Δo < P (Pairing Energy): Weak field ligand (e.g., F^- F−, Cl^- Cl−). Electrons enter e_g eg before pairing. Forms **High Spin** complex.
  - If $\Delta_o > P$Δo > P: Strong field ligand (e.g., CN^- CN−, COCO). Electrons pair up in t_{2g} t2g before entering e_g eg. Forms **Low Spin** complex.

D. Isomerism Types

1. **Linkage Isomerism**: Arises with **ambidentate ligands** (ligands having two donor atoms): e.g., $[Co(NH_3)_5(NO_2)]Cl_2$ [Co(NH3)5(NO2)]Cl2 (nitro, yellow) and $[Co(NH_3)_5(ONO)]Cl_2$ [Co(NH3)5

(ONO)]Cl2 (nitrito, red).

2. **Ionization Isomerism**: Exchange of ions inside and outside coordination sphere: e.g., $[Co(NH_3)_5SO_4]Br$[Co(NH3)5SO4]Br (gives $AgBr$AgBr ppt with $AgNO_3$ AgNO3) and $[Co(NH_3)_5Br]SO_4$ [Co(NH3)5Br]SO4 (gives $BaSO_4$ BaSO4 ppt with $BaCl_2$ BaCl2).
3. **Coordination Isomerism**: Exchange of ligands between cationic and anionic entities: $[Co(NH_3)_6][Cr(CN)_6]$[Co(NH3)6][Cr(CN)6] and $[Cr(NH_3)_6][Co(CN)_6]$[Cr(NH3)6][Co(CN)6].
4. **Hydrate (Solvate) Isomerism**: $[Cr(H_2O)_6]Cl_3$ [Cr(H2O)6]Cl3 (violet) vs $[Cr(H_2O)_5Cl]Cl_2 \cdot H_2O$[Cr(H2O)5Cl]Cl2 · H2O (grey-green).
5. **Optical Isomerism**: Non-superimposable mirror images (chiral complexes): e.g., $[M(en)_3]^{3+}$ [M(en)3]3+ exists as dd- and ll-forms.

PART 3: ORGANIC CHEMISTRY (33 / 33 MARKS)

Organic Chemistry makes up nearly half the paper. It is divided into 4 Pillars: Named Reactions, Reaction Mechanisms, Chemical Distinguishing Tests, and Scientific Reasoning.

3.1 The 15 Non-Negotiable Named Reactions

1. **Aldol Condensation**:
   - Condition: Aldehydes or ketones having **at least one αα-hydrogen** in presence of dilute alkali (dil. $NaOH$dil. NaOH).

     $$2CH_3CHO \xrightarrow{\text{dil. NaOH}} CH_3-CH(OH)-CH_2-CHO \xrightarrow{\Delta,\ -H_2O} CH_3-CH=CH-CHO \text{ (But-2-enal)}$$

     2CH3CHO dil. NaOH

     CH3 − CH(OH) − CH2 − CHO Δ,−H2O

     CH3 − CH = CH − CHO (But-2-enal)
2. **Cannizzaro Reaction**:
   - Condition: Aldehydes having **no αα-hydrogen** ($HCHO$HCHO, C_6H_5CHOC6H5CHO) undergo self oxidation-reduction (disproportionation) in concentrated (50%50%) alkali.

     $$2HCHO + \text{conc. } NaOH \rightarrow HCOONa \text{ (Sodium formate)} + CH_3OH \text{ (Methanol)}$$

     2HCHO + conc. NaOH → HCOONa (Sodium formate) + CH3OH (Methanol)

     $$2C_6H_5CHO + \text{conc. } KOH \rightarrow C_6H_5COOK + C_6H_5CH_2OH \text{ (Benzyl alcohol)}$$

     2C6H5CHO + conc. KOH → C6H5COOK + C6H5CH2OH (Benzyl alcohol)
3. **Reimer-Tiemann Reaction**:
   - Phenol reacts with Chloroform in aqueous $NaOH$NaOH to form Salicylaldehyde.

     $$C_6H_5OH + CHCl_3 + 3NaOH \xrightarrow{340\ K} o\text{-hydroxybenzaldehyde (Salicylaldehyde)} + 3NaCl + 2H_2O$$

C6H5OH + CHCl3 + 3NaOH 340 K

o-hydroxybenzaldehyde (Salicylaldehyde) + 3NaCl + 2H2O

4. **Kolbe's Reaction**:
   - Sodium phenoxide treated with CO_2 CO2 at $400\ K$400 K and $4-7\ atm$4 − 7 atm, followed by acidification, yields Salicylic acid.

     $$C_6H_5ONa + CO_2 \xrightarrow{400\,K,\,4-7\,atm} \text{Sodium salicylate} \xrightarrow{H^+} \text{Salicylic acid (2-hydroxybenzoic acid)}$$

     C6H5ONa + CO2 400 K, 4−7 atm

     Sodium salicylate H+

     Salicylic acid (2-hydroxybenzoic acid)

5. **Williamson Ether Synthesis**:
   - Reaction of alkyl halide with sodium alkoxide: $R-X+R'-O^-Na^+ \rightarrow R-O-R'+NaX$R − X + R′ − O−Na+ → R − O − R′ + NaX.
   - *Crucial Rule for Hard Papers*: The alkyl halide must be **primary (**1° **1∘)**. If a 3° 3∘ alkyl halide is taken, **elimination occurs to yield an alkene** instead of an ether because alkoxide acts as a strong base.

6. **Clemmensen Reduction**:
   - Carbonyl group ($>C=O$> C = O) reduced to methylene ($>CH_2$ > CH2) using **Zinc Amalgam and concentrated** HCl**HCl (**Zn-$Hg/conc.\ HCl$**Zn-Hg/conc. HCl)**.

     $$CH_3COCH_3 + 4[H] \xrightarrow{Zn\text{-}} CH_3CH_2CH_3\ (\text{Propane}) + H_2O$$

     CH3COCH3 + 4[H] Zn-Hg/conc. HCl

     CH3CH2CH3 (Propane) + H2O

7. **Wolff-Kishner Reduction**:
   - Carbonyl group reduced to methylene using **Hydrazine (**NH_2NH_2 **NH2NH2) followed by heating with** KOH**KOH in ethylene glycol**.

     $$>C=O \xrightarrow{NH_2NH_2,\,-H_2O} >C=N-NH_2 \xrightarrow{KOH\,/\,glycol,\,\Delta} >CH_2 + N_2\uparrow$$

     > C = O NH2NH2,−H2O

     > C = N − NH2 KOH / glycol, Δ

     > CH2 + N2 ↑

8. **Rosenmund Reduction**:
   - Acyl chloride hydrogenated over Palladium supported on Barium sulphate poisoned with sulphur/quinoline (Pd-$BaSO_4$ Pd-BaSO4) to yield Aldehyde.

     $$R-COCl + H_2 \xrightarrow{Pd\text{-}} R-CHO + HCl$$

     R − COCl + H2 Pd-BaSO4

R − CHO + HCl

9. **Stephen Reaction**:
   - Nitriles reduced with stannous chloride and HClHCl ($SnCl_2$ /HClSnCl2/HCl), followed by hydrolysis, gives aldehyde.

$$R-CN+2[H]+HCl \xrightarrow{SnCl_2/HCl} R-CH=NH\cdot HCl \xrightarrow{H_3O^+} R-CHO+NH_4Cl$$

R − CN + 2[H] + HCl SnCl2/HCl

R − CH = NH · HCl H3O+

R − CHO + NH4Cl

10. **Etard Reaction**:
    - Toluene oxidized by chromyl chloride (CrO_2Cl_2 CrO2Cl2) in CS_2 CS2 to form a chromium complex, which on hydrolysis yields Benzaldehyde.

$$C_6H_5CH_3+2CrO_2Cl_2 \xrightarrow{CS_2} \text{Chromium complex} \xrightarrow{H_3O^+} C_6H_5CHO$$

C6H5CH3 + 2CrO2Cl2 CS2

Chromium complex H3O+

C6H5CHO

11. **Hell-Volhard-Zelinsky (HVZ) Reaction**:
    - Carboxylic acids having an αα-hydrogen react with Chlorine or Bromine in presence of **red Phosphorus** to give αα-halocarboxylic acids.

$$CH_3COOH+Cl_2 \xrightarrow{Red\ P} CH_2Cl\text{-}COOH+HCl$$

CH3COOH + Cl2 Red P

CH2Cl-COOH + HCl

12. **Hoffmann Bromamide Degradation Reaction**:
    - Primary amide treated with Bromine and ethanolic/aqueous NaOHNaOH yields primary amine containing **one carbon atom less than the parent amide**.

$$R-CONH_2+Br_2+4NaOH \rightarrow R-NH_2+Na_2CO_3+2NaBr+2H_2O$$

R − CONH2 + Br2 + 4NaOH → R − NH2 + Na2CO3 + 2NaBr + 2H2O

13. **Carbylamine Reaction (Isocyanide Test)**:
    - Primary aliphatic or aromatic amine heated with Chloroform and ethanolic KOHKOH produces an extremely **foul-smelling isocyanide (carbylamine)**.

$$R-NH_2+CHCl_3+3KOH(alc) \xrightarrow{\Delta} R-NC\ (\text{Foul smell})+3KCl+3H_2O$$

R − NH2 + CHCl3 + 3KOH(alc) Δ

R − NC (Foul smell) + 3KCl + 3H2O

14. **Sandmeyer Reaction**:
    - Diazonium salt treated with $CuCl/HCl$CuCl/HCl, $CuBr/HBr$CuBr/HBr, or $CuCN/KCN$CuCN/KCN to form Chlorobenzene, Bromobenzene, or Benzonitrile.

$$C_6H_5N_2^+Cl^- \xrightarrow{CuCl/HCl} C_6H_5Cl + N_2 \uparrow$$

C6H5N2+Cl− CuCl/HCl

C6H5Cl + N2 ↑

15. **Azo Coupling Reaction**:
    - Benzene diazonium chloride reacts with Phenol in mildly basic medium ($pH\ 9-10$pH 9 − 10) or Aniline in mildly acidic medium ($pH\ 4-5$pH 4 − 5) to form intensely colored azo dyes.

$$C_6H_5N_2^+Cl^- + C_6H_5OH \xrightarrow{OH^-,\ 273-278\ K} p\text{-hydroxyazobenzene (Orange dye)} + Cl^- + H_2O$$

C6H5N2+Cl− + C6H5OH OH−,273−278 K

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p-hydroxyazobenzene (Orange dye) + Cl− + H2O

3.2 The 5 Mandatory Reaction Mechanisms

Mechanism 1: S_N1SN1 vs S_N2SN2 Mechanisms (Haloalkanes)

Parameter	S_N1SN1 (Substitution Nucleophilic Unimolecular)	S_N2SN2 (Substitution Nucleophilic Bimolecular)
Kinetics & Order	First order: $\text{Rate} = k[R-X]$Rate = k[R − X]	Second order: $\text{Rate} = k[R-X][Nu^-]$ Rate = k[R − X][Nu−]
Number of Steps	**Two steps**: Step 1: Slow ionization to form carbocation intermediate; Step 2: Fast attack by nucleophile.	**Single concerted step**: Simultaneous bond breaking and bond making through a pentacoordinated **Transition State**.
Reactivity Order	$3^\circ > 2^\circ > 1^\circ > CH_3X$3° > 2° > 1° > CH3X (Governed by stability of carbocation: $3^\circ > 2^\circ > 1^\circ$3° > 2° > 1°).	$CH_3X > 1^\circ > 2^\circ > 3^\circ$CH3X > 1° > 2° > 3° (Governed by **steric hindrance**; bulky groups block attack).
Stereochemistry	**Racemization** (Carbocation is planar sp^2 sp2; attack occurs with equal probability from front and back).	**Complete Inversion of Configuration** (**Walden Inversion**; backside attack opposite to leaving group).
Solvent Favored	Polar protic solvents (H_2OH2O, alcohol, acetic acid stabilize ions).	Polar aprotic solvents (Acetone, DMSO, DMF do not solvate nucleophile).

Mechanism 2: Acid-Catalyzed Dehydration of Ethanol

- **At** $443\ K$**443 K (Forms Ethene)**:

- *Step 1: Protonation of alcohol*: $CH_3CH_2OH + H^+ \rightleftharpoons CH_3CH_2OH_2^+$ CH3CH2OH + H+ ⇌ CH3CH2 OH2+ (Protonated alcohol / oxonium ion).
- *Step 2: Formation of carbocation (Slow, Rate determining step)*: $CH_3CH_2OH_2^+ \xrightarrow{slow} CH_3CH_2^+ + H_2O$ CH3CH2OH2+ slow

  CH3CH2+ + H2O.
- *Step 3: Elimination of proton*: $CH_3CH_2^+ \rightarrow CH_2 = CH_2 + H^+$ CH3CH2+ → CH2 = CH2 + H+.

- **At** $413\ K$**413 K with excess ethanol (Forms Diethyl Ether)**:
  - Carbocation formed in step 2 is attacked by a second neutral ethanol molecule by S_N2SN2 pathway to yield protonated ether:
    $CH_3CH_2^+ + CH_3CH_2OH \rightarrow CH_3CH_2 - O^+(H) - CH_2CH_3 \xrightarrow{-H^+} CH_3CH_2 - O - CH_2CH_3$ CH3CH2+ + CH3CH2OH → CH3CH2 − O+(H) − CH2CH3 −H+

    CH3CH2 − O − CH2CH3.

Mechanism 3: Acid-Catalyzed Hydration of Alkenes to Alcohols

- **Markovnikov Addition**:
  - *Step 1: Electrophilic attack of* H_3O^+ *H3O+ on double bond*: $H_2O + H^+ \rightleftharpoons H_3O^+$ H2O + H+ ⇌ H3 O+; $CH_3 - CH = CH_2 + H^+ \xrightarrow{slow} CH_3 - CH^+ - CH_3$ CH3 − CH = CH2 + H+ slow

    CH3 − CH+ − CH3 (2° 2∘ carbocation formed preferentially over 1° 1∘).
  - *Step 2: Nucleophilic attack of water on carbocation*:
    $CH_3 - CH^+ - CH_3 + H_2O \rightleftharpoons CH_3 - CH(OH_2^+) - CH_3$ CH3 − CH+ − CH3 + H2O ⇌ CH3 − CH(OH2+) − CH3.
  - *Step 3: Deprotonation*:
    $CH_3 - CH(OH_2^+) - CH_3 + H_2O \rightarrow CH_3 - CH(OH) - CH_3$ (Propan-2-ol) $+ H_3O^+$ CH3 − CH(OH2+) − CH3 + H2O → CH3 − CH(OH) − CH3 (Propan-2-ol) + H3O+.

Mechanism 4: Nucleophilic Addition to Carbonyl Groups ($>C=O$> C = O)

- Carbonyl carbon is sp^2 sp2 hybridized and coplanar. Due to higher electronegativity of oxygen, the carbon develops a partial positive charge (δ^+ δ+).
- *Step 1*: Nucleophile (Nu^- Nu−) attacks the electrophilic carbonyl carbon perpendicularly to the plane, converting sp^2 sp2 planar carbon into an sp^3 sp3 **tetrahedral alkoxide intermediate**.
- *Step 2*: The tetrahedral intermediate captures a proton (H^+ H+) from the medium to yield the neutral addition product.

3.3 The Master Table of Chemical Distinguishing Tests

Pair of Compounds	Reagent / Test	Observation for Compound A	Observation for Compound B
$1^\circ, 2^\circ, 3^\circ$ **1∘, 2∘, 3∘ Alcohols**	**Lucas Reagent** ($\text{anhydrous } ZnCl_2 + \text{conc. } HCl$ anhydrous ZnCl2 + conc. HCl)	3° 3∘ alcohol: Turbidity appears **immediately**; 2° 2∘ alcohol: Turbidity appears in **5 minutes**	1° 1∘ alcohol: **No turbidity at room temperature** (appears only on heating)
Aldehyde vs Ketone	**Tollen's Reagent** (Ammoniacal $AgNO_3$ AgNO3)	Aldehyde: **Silver mirror** forms on test tube walls	Ketone: **No reaction**
	Fehling's Solution ($\text{Fehling A}+\text{Fehling B}$ Fehling A + Fehling B)	Aliphatic Aldehyde: **Red precipitate** of Cu_2O Cu2 O	Ketone / Benzaldehyde: **No precipitate**
Methyl Ketone vs Other Ketone (e.g., Pentan-2-one vs Pentan-3-one)	**Iodoform Test** ($I_2 + NaOH$ I2 + NaOH)	Pentan-2-one (has CH_3CO- CH3CO− group): **Yellow precipitate of** CHI_3 **CHI3**	Pentan-3-one: **No yellow precipitate**
Ethanol vs Methanol	**Iodoform Test** ($I_2 + NaOH$ I2 + NaOH)	Ethanol (has $CH_3CH(OH)-$ CH3 CH(OH)−): **Yellow precipitate**	Methanol: **No yellow precipitate**
Phenol vs Carboxylic Acid	$NaHCO_3$ **NaHCO3 (Sodium Bicarbonate)**	Carboxylic acid: **Brisk effervescence of** CO_2 **CO2 gas**	Phenol: **No effervescence** (Phenol is too weak to decompose $NaHCO_3$ NaHCO3)
	Neutral $FeCl_3$ **FeCl3 Solution**	Phenol: Gives **violet coloration**	Carboxylic acid: **Buff colored precipitate** or no violet color
1° **1∘ Amine vs** $2^\circ/3^\circ$ **2∘/3∘ Amine**	**Carbylamine Test** ($CHCl_3 + \text{alc. KOH}$ CHCl3 + alc. KOH, heat)	1° 1∘ amine (Aliphatic or Aromatic): **Extremely offensive foul smell of isocyanide**	2° 2∘ and 3° 3∘ amines: **No foul smell**
$1^\circ, 2^\circ, 3^\circ$ **1∘, 2∘, 3∘ Amines**	**Hinsberg Reagent** (Benzenesulphonyl chloride)	1° 1∘ Amine: Forms precipitate **soluble in** KOH **KOH**	2° 2∘ Amine: Forms precipitate **insoluble in** KOH **KOH**; 3° 3∘ Amine: **Does not react**
Aniline vs Ethylamine	**Azo Dye Test** ($NaNO_2 + HCl$ NaNO2 + HCl at $0-5^\circ C$ 0 − 5∘C, then alkaline ββ-naphthol)	Aniline: **Bright orange/scarlet red dye** forms	Ethylamine: **Bubbles of** N_2 **N2 gas evolved**, no dye

3.4 Essential Organic Reasoning Questions

1. **Why are haloarenes less reactive towards nucleophilic substitution reactions than haloalkanes?**
   Answer: (i) **Resonance effect**: C—Cl bond acquires partial double bond character, making it shorter and stronger. (ii) **Difference in hybridization of carbon**: Carbon in haloarenes is sp^2 sp2 hybridized (more electronegative, holds electrons tighter) vs sp^3 sp3 in haloalkanes. (iii) **Instability of phenyl cation**: Self-ionization to phenyl cation is impossible. (iv) **Repulsion**: Nucleophile is repelled by electron-rich benzene ring.
2. **Order of basicity of aliphatic amines in aqueous solution**:
   Answer:
   - For Methyl substituted amines: $(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N > NH_3$ (CH3)2NH > CH3NH2 > (CH3)3N > NH3 ($2^\circ > 1^\circ > 3^\circ > \mathrm{NH_3}$ 2∘ > 1∘ > 3∘ > NH3).
   - For Ethyl substituted amines: $(C_2H_5)_2NH > (C_2H_5)_3N > C_2H_5NH_2 > NH_3$ (C2H5)2NH > (C2H5)3N > C2H5NH2 > NH3 ($2^\circ > 3^\circ > 1^\circ > \mathrm{NH_3}$ 2∘ > 3∘ > 1∘ > NH3).
     Reason: Combined balance of **Inductive effect (+I)**, **Solvation effect (H-bonding with water)**, and **Steric hindrance**. In gaseous phase, only +I effect works: $3^\circ > 2^\circ > 1^\circ > \mathrm{NH_3}$ 3∘ > 2∘ > 1∘ > NH3.
3. **Why is Aniline less basic than aliphatic amines and ammonia?**
   Answer: The lone pair of electrons on the nitrogen atom of aniline is delocalized over the aromatic benzene ring through resonance, making it less available for protonation.
4. **Acidity comparison: Carboxylic Acids vs Phenols vs Alcohols**:
   Order: $\mathrm{Carboxylic\ Acid > Phenol > Water > Alcohols}$Carboxylic Acid > Phenol > Water > Alcohols.
   Reason: Carboxylate ion ($\mathrm{RCOO^-}$ RCOO−) is stabilized by **two equivalent resonance structures** with negative charge on electronegative oxygen atoms. Phenoxide ion has negative charge delocalized over less electronegative carbon atoms. Alkoxide ion ($\mathrm{RO^-}$ RO−) has no resonance and is destabilized by electron-donating alkyl group (+I effect).
   (EWG like $-\mathrm{NO_2}$ −NO2 increases acidity by stabilizing the conjugate base; EDG like $-\mathrm{CH_3}$ −CH3 decreases acidity).
5. **Why is ortho-nitrophenol steam volatile while para-nitrophenol is not?**
   Answer: Ortho-nitrophenol has **intramolecular hydrogen bonding** (within the same molecule), resulting in a discrete chelate ring and lower boiling point. Para-nitrophenol has **intermolecular hydrogen bonding** between adjacent molecules, causing molecular association and a significantly higher boiling point.

3.5 Unit 10: Biomolecules (7 Marks)

1. **Reducing vs Non-Reducing Sugars**:
   - *Reducing*: Carbohydrates that reduce Fehling's solution and Tollen's reagent due to presence of free aldehydic or ketonic group (e.g., Glucose, Fructose, Maltose, Lactose).

- *Non-reducing*: Carbohydrates where reducing groups (aldehydic/ketonic) are locked in **glycosidic linkage** (e.g., **Sucrose**; formed between C-1 of αα-glucose and C-2 of ββ-fructose).

2. **What is Invert Sugar?**
   Answer: Sucrose is dextrorotatory ($+66.5^\circ$ +66.5∘). On acid hydrolysis, it produces an equimolar mixture of D-(+)D-(+)-glucose ($+52.5^\circ$ +52.5∘) and D-(-)D-(-)-fructose (-92.4° −92.4∘). Because the laevorotation of fructose is greater than the dextrorotation of glucose, the resulting mixture is **laevorotatory**. This change in sign of optical rotation from dextro to laevo is called **inversion of sugar**, and the product mixture is called **invert sugar**.
3. **Chemical Reactions of D-Glucose Proving its Structure**:
   - Reaction with HIHI, heat $\rightarrow$ → n**n-Hexane** (proves a continuous 6-carbon straight chain).
   - Reaction with NH_2OHNH2OH (Hydroxylamine) $\rightarrow$ → **Oxime** (proves presence of a carbonyl group).
   - Reaction with HCNHCN $\rightarrow$ → **Cyanohydrin** (confirms carbonyl group).
   - Reaction with Bromine water (Br_2/H_2OBr2/H2O, mild oxidizer) $\rightarrow$ → **Gluconic acid** (proves carbonyl is an **aldehyde group** $-CHO$−CHO).
   - Reaction with Acetic Anhydride $\rightarrow$ → **Glucose pentaacetate** (proves presence of **5 —OH groups** on different carbons).
   - Reaction with Conc. HNO_3 HNO3 $\rightarrow$ → **Saccharic acid** (proves presence of a primary alcohol group $-CH_2OH$−CH2OH).
4. **Proteins & Denaturation**:
   - **Peptide Linkage**: An amide bond ($-CO-NH-$ −CO − NH−) formed between the $-COOH$ −COOH group of one amino acid and the $-NH_2$ −NH2 group of another with loss of a water molecule.
   - **Denaturation of Proteins**: When a protein in its native state is subjected to physical change (heat) or chemical change (change in pH), the hydrogen bonds are broken. The globules unfold and helices uncoil, causing the protein to lose its biological activity.
     Crucial Fact: During denaturation, **secondary and tertiary structures are destroyed, but the primary structure (peptide sequence) remains completely intact**. (Examples: Coagulation of egg white on boiling; curdling of milk).
5. **Differences between DNA and RNA**:

Characteristic	DNA	RNA
Sugar Present	ββ-D-2-deoxyribose	ββ-D-ribose
Nitrogenous Bases	Adenine, Guanine, Cytosine, **Thymine (T)**	Adenine, Guanine, Cytosine, **Uracil (U)**
Structure	Double-stranded helical structure	Single-stranded structure
Function	Stores and transmits genetic information	Controls protein synthesis

6. **Vitamins Classification**:
    - **Fat Soluble**: Vitamins **A, D, E, K** (Stored in liver and adipose tissue).
    - **Water Soluble**: Vitamins **B-complex and C** (Excreted readily in urine, cannot be stored in body, require regular dietary intake; Exception: Vitamin B_{12} B12).
    - *Deficiency Diseases*: Vitamin A $\rightarrow$ → Night blindness / Xerophthalmia; Vitamin B_1 B1 $\rightarrow$ → Beri-beri; Vitamin C $\rightarrow$ → Scurvy (bleeding gums); Vitamin D $\rightarrow$ → Rickets; Vitamin K $\rightarrow$ → Increased blood clotting time.
7. **Zwitterion**: An internal salt structure formed in amino acids when the acidic carboxyl group ($-COOH$ −COOH) transfers a proton to the basic amino group ($-NH_2$ −NH2), producing a dipolar dipolar ion with net neutral charge: $H_3N^+ - CH(R) - COO^-$ H3N+ − CH(R) − COO−.

PART 4: THE 5-MARK LONG ANSWER PREDICTIONS (SECTION D)

Section D carries **3 Questions** $\times$ **× 5 Marks = 15 Marks** (with 100% internal choice). In JKBOSE, these 3 questions follow a rigid distribution:

1. **Question 20 (Physical Chemistry Choice)**:
    - **Option A**: Integrated Rate Law for First-Order Reaction (Derivation + Half-life proof + standard $t_{99\%} = 2t_{90\%}$ t99% = 2t90% numerical).
    - *OR* **Option B**: Electrochemistry (Kohlrausch's Law statement + 2 applications + Nernst equation numerical with EMF & ΔG° ΔG∘).
2. **Question 21 (Inorganic / Physical Choice)**:
    - **Option A**: Solutions (Derivation of ΔT_f ΔTf or ΔT_b ΔTb + Van 't Hoff factor relation with αα + abnormal molar mass numerical).
    - *OR* **Option B**: Coordination Compounds & d-Block (VBT explanation of $[Fe(CN)_6]^{3-}$ [Fe(CN)6]3− vs $[FeF_6]^{3-}$ [FeF6]3− + IUPAC names + Lanthanoid contraction consequences).
3. **Question 22 (Organic Chemistry Choice)**:
    - **Option A**: Aldehydes & Ketones (Aldol condensation + Cannizzaro reaction + Clemmensen/Wolff-Kishner reduction + chemical test to distinguish ethanal and propanal).
    - *OR* **Option B**: Alcohols, Phenols & Amines (Acid-catalyzed dehydration mechanism of ethanol at $443\ K$443 K + Reimer-Tiemann & Kolbe's reactions + Basicity order of amines in aqueous solution).

5. Exam Execution Strategy (How to prevent silly mistakes)

1. **Never write bare numbers in Physical Numericals**: Every value must end with its exact SI unit ($\text{mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1}$ mol · L−1 · s−1, $\text{S} \cdot \text{cm}^2 \cdot \text{mol}^{-1}$ S · cm2 · mol−1, $\text{g} \cdot \text{mol}^{-1}$ g · mol−1, VV, $\text{J} \cdot \text{mol}^{-1}$ J · mol−1). Board paper setters penalize 0.5 to 1 mark for missing units.
2. **Include Reaction Conditions in Organic**: When writing named reactions, always specify reagents above the arrow (dil. NaOHdil. NaOH, Zn-Hg / conc. HClZn-Hg / conc. HCl, $443\ \text{K}$443 K, H^+ H+).
3. **Step Marking in Derivations**: Always state the starting equation, write the integration boundary conditions, and highlight the final formula in a box.